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29 April 1955
SUPERSEDING
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15 September 1951
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MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING, ELECTRIC MOTOR DRIVEN, NAVAL SHIPBOARD DIESEL ENGINE AIR STARTING SERVICE

1. SCOPE

1.1 Scope.- This specification covers class T motor driven air compressors, for Diesel engine air starting installations aboard Naval ships.

1.2 Classification.- Diesel starting air compressors shall be of the following sizes as specified (see 6. 1):

Size 10 - Not less than 10 cf. m, of free air per minute at 600 p. s. i. g. discharge pressure.
Size 50 - Not less than 50 c.f. m. of free air per minute at 600 p. s. i. g. discharge pressure.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearing, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical, and Bearings, Roller, SeLf-Aligning,
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
QQ-A-561 - Aluminum Alloy 2S; Plate and Sheet.
QQ-C-576 - Copper Plates, Sawed Bars, Sheets, and Strips.
QQ-I-652 - Iron Castings, Gray,
QQ-S-763 - Steel, Corrosion-Resisting: Bars and Forgings (Except for Reforging).
QQ-S-766 - Steel, Corrosion-Resisting, Plates, Sheets, Strips and Structural Shapes.
GGG-P-781 - Pullers, Bushing, Bearing, Gear, Wheel Hub and Cylinder Sleeve Installing and Removing.
PPP- B-601 - Boxes, Wood, Cleated-Plywood.

MILITARY

JAN- P-100 - Packaging and Packing for Overseas Shipment: General Specification.
JAN-P-104 - Packaging and Packing for Overseas Shipment Crates, Sheathed, Wood, Nailed.
MIL-P-116 - Preservation, Methods of,
JAN-P-121 - Barrier-Materials, Greaseproof.

*(See 6. 4,)

FED SUP CLASS
4310

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JAN-P-127 - Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant.
MIL-B-233 - Boxes, Repair Parts.
JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-G-858 - Gray Iron Alloy: Castings (Scale-Resisting).
MIL-S-867 - Steel Castings, Corrosion-Resisting Austenitic.
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat-Treated).
MIL-B-892 - Bronze, Phosphor, Rolled or Drawn; Ears, Plates, Rods, Sheets, and Strips.
MIL-N-894 - Nickel Copper Alloy Bars, Plates, and Other Wrought Forms,
MI L-S-901 - Shockproof Equipment, Class HI (High-lm.met), Shipboard Application, Tests for,
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-A-958 - Aluminum Base Alloy, Permanent Mold Castings (Marine Service),
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-B-994 - Brass, Naval, Wrought.
JAN-S- 1160- Steel, Spring, Round or Square (Including Rectangular Sections Having Thicknesses Greater Than Half Their Width).
MIL- T-1368 - Tubing, Nickel Copper Alloy, Seamless and Welded.
MIL-C-2174 - Controllers and Master Switches, Direct-Current (Naval Shipboard Use).
MIL-C-2212 - Controllers and Master Switches, Alternating-Current (Naval Shipboard Use).
MIL-C-3769 - Crates, Inter mediate, Sheathed, Wood, Nailed (for Maximum Net Loads of 3000 Pounds).
MIL-T-15005 - Tubes, Copper Nickel Alloy (Condenser and Heat Exchanger).
MIL-I- 15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-T-15071 - Technical Manuals for Mechanical and Electrical Equipment.
MIL-S-15083 - Steel; Castings,
MIL-P-15130 - Paint., Outside, Haze-Gray, No. 27 (Formula No. 5H) for Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechnical Equipment (Naval Shipboard Use).
MIL-C-15726 - Copper Nickel Alloy: Bars, Plates, Ross, Sheets, and Strips.
MIL-C-15730 - Coolers, Fluid, Industrial; Lubricating Oil and Fresh Water, (Naval Shipboard Use).
MIL-B-15939 - Bronze, Aluminum: Wrought.
MIL-P-16018 - Packaging of Switchboards, Controllers and Motor Star ters (Over 40 Pounds).
MIL-B-16033 - Bronze, Aluminum: Castings.
MIL-S-16113 - Steel, Plate, Hull and Ordnance, Structural, Black (Uncoated) and Zinc Coated (Galvanized),
MIL-B- 16261- Bronze, Bear ings: Castings.
MIL-P-16298 - Preservation, Packaging, Packing and Marking of Electric Machines Having Rotating Parts (Includes Associated Repair Parts).
MIL-T-16420 - Tube, Copper Nickel (70-30 and 90-10), Seamless and Welded.
MIL-B-16540 - Bronze, Phosphor: Castings.
MIL-B-16541 - Bronze, Valves: Castings.

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MILITARY (cont'd.)

MIL-M-16576 - Metal, Gun: Castings.
MIL-M- 17060 - Motors, Alternating Current, Integral HP (Shipboard Use).
MIL-I-17166 - iron Castings: Nodular Graphitic (Ductile Iron) (for Shipboard Applications).
MIL-I-17214 - Indicators, Permeability; Low-Mu (Go-No-Go).
MIL- L-17331 - Lubricating Oil, Turbine (Noncorrosive).
MIL-M-17413 - Motor, Direct-Current, Integral HP (Shipboard Use).
MIL-S-17509 - Steel Castings, Austenitic, Chromium-Nickel, Low Magnetic Permeability.
MIL-P-17652 - Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Parts.
MIL-S- 17906- Steel Sheet and Other Wrought Forms, Corrosion Resisting, Austenitic (Low Magnetic Permeability), Chromium-Manganese-Nickel.
MIL-A-18001 - Anodes, Corrosion Preventive, Plates, Slabs, Discs, Rods, Zinc, Cast in Core.
MIL-S- 20166- Steel: Bars and Shapes (for Drop and Miscellaneous Construction) (Including Material for Hull).

NAVY DEPARTMENT

General Specifications for Inspection of Material.
45G1 - Gages, Pressure and Vacuum (for Air, Ammonia, Oil, Steam, and Water).
General Specifications for Machinery:
Section S1 -4 - Welding and Allied Processes.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS

B-214 - Root Connections for Attached Piping.
5000-S4823-841055 - Check off List for Zincs in Machinery Installation, Instructions for Preparation of Form.
5000 -S4823-841337 - Unions, Silver Alloy Brazing in Machinery Installations, Instruction for Preparation of Form.

PUBLICATIONS

MILITARY

JPI- 9- Packaging and Packing of Machine Tools and Accessory Parts for Overseas Shipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agencey or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

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NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN TRUCKING ASSOCIATION
Motor Freight Classification Rules,

(Application for copies should be addressed to the American Trucking Association. 1424 16th St. N.W. , Washington 6, D. C.)

COMPRESSED AIR HANDBOOK
Second Edition

(Application for copies should be addressed to the McGraw and Hill Publishing Company, New York City, New York.)

3. REQUIREMENTS

3.1 Qualification.- The compressors furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Description.-

3. 2.1 Class T compressors shall be reciprocating, vertical, single acting, multi-stage machines having capacities of 10 c. f. m. or 50 cf. m. of free air per minute actually delivered at 600 pounds per square inch gage (p.s.i.g.) discharge pressure. Each compressor shall be capable of continuous operation at any discharge pressure from 150 to 600 p. s. i. g.

3.2.2 A complete compressor unit shall consist of the compressor and motor mounted on a common bedplate, and shall include all necessary appurtenances and electrical control equipment.

3.2.3 Safety features including guards for belts, flywheels, sheaves, and other exposed moving parts, shall be incorporated in the compressor design.

3.2.4 The drive shall be of the multi- "V"-belt type. Suitable means shall be provided for tightening and adjusting the belts.

3.3 General design. -

3.3. 1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain air compressors of such design that they will operate over a long period of years with a minimum of servicing. Where wear or corrosion is unavoidable the parts subjected to such wear or corrosion shall be of the best materials available for the purpose in order to reduce these detrimental effects to a minimum. The design and construction of all air compressors shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion.
(d) Economy.

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(e) Satisfactory operation when inclined as follows:
(1) Up to 15 degrees in any direction (permanently inclined).
(2) With the ship rolling up to 45 degrees from the vertical to either side,
(3) With the pitching 10 degrees up and down from the normal horizontal plane.
(f) Minimum space and weight.

3.3.2 Unless otherwise specified in the contract or order the following design conditions shall be assumed:

(a) Ambient and intake air temperature - 122° Fahrenheit (F.),
(b) Intake air pressure - 14.7 p.s.i. absolute.
(c) Relative humidity - 50 percent,.
(d) Maximum permissible temperature of air leaving the aftercooler - 100° F.
(e) Temperature of cooling w ater at inlet - 85° F,
(f) Kind of cooling water - sea (salt).

3, 3.3 Piston speed. - The piston speed shall not exceed 800 feet per minute.

3. 3.4 Gage board, drain valves and all features of an operational nature shall be located on or operable from one side, designated as the operating side of the compressor, so as to be accessible if the compressor is installed adjacent to a bulkhead.

3.2.5 The crank case, frame, housing, cylinders, and other fixed parts or attached components of the assembly shall be rigidly secured and held in alignment by dowel pins, fitted bolts, keys, rab beted or tongue and groove joints or other suitable means.

3. 3.6 Air cylinders and heads. - The cylinders and heads shall be provided with ample water-jackets (see 3. 3.15. 1. 1). So far as practicable hand-holes shall be provided for access to the jacket spaces. Cylinder liners shall be held rigidly in place. The liners shall be counterbored slightly at the ends to permit the piston rings to override the working bore to avoid the formation of shoulders. The design and machining of cylinders and liners shall be such that the complete piston assembly can be removed and replaced. To this end all shoulders where piston rings may catch shall be avoided and liners shall be properly beveled,

3. 3. 7 Connecting rods. - The lower end of connecting rods shall be arranged for renewal of crankpin bearings which shall be of thin shell precision type, Bolts and nuts for securing crankpin bearings shall be positively locked, The upper end of connecting *rods* connected to crossheads or pistons shall be fitted with renewable bearings, Provision for adjustment shall be made as required for proper operation and maintenance of the compressor. Any take-up bolts and nuts shall be positively locked. Upper end of connecting rods snail be of one-piece construction.

3. 3.8 Crankshaft. - The crankshaft shall be a *single* forging. The cranks shall be arranged to give the best possible running balance. Counterweights shall be fitted if necessary. The design of the shaft shall be such as to permit convenient use of a portable tachometer.

3. 3.9 Flywheels. - Each compressor shall be equipped with a flywheel of sufficient size to provide smooth running of the compressor and motor. Flywheels shall be keyed to the crankshaft and shall be pressed or securely clamped thereon. A separate flywheel will not be required if ample flywheel effect is provided by the sheave.

3.3.10 Main and thrust bearings. - Main bearings shall be readily renewable, The main bearing shall preferably be of the tapered roller bearing type in accordance with Specification FF-B-185 which shall be so arranged as to *take* up the thrust due to roll and pitch of the vessel. Consideration will be given to the use of deep groove or angular contact type ball bearings in accordance with Specification FF-B-171.

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3.3.11 Lubrication.- An adequa te and self-contalneci lubricating system of the force feed type shall be provided for the main bearings, crankpin bearings, wrist-pin bearings, gears, pinion and similar items. Consideration will be given to splash feed systems or a combination of splash and force feed systems, The crankpit shall be suitably housed to form a reservoir for the lubricating oil, Splash guards shall be fitted to prevent oil from the crankcase entering the air cylinders. The crankpit reservoir shall be provided with suitable fittings for filling and draining of the lubricating oil. A suitable oil-level gage, marked to show the high, low and normal oil levels, shall be fitted. Gage glasses, if used, shall be renewable without draining the Oil from the reservior. Oil shall be drawn from the reservoir by one or more pumps, positively driven from the compressor crankshaft. The arrange ment of the reservior and pump connection shall be such that oil will not be lost from th reservoir, nor the oil pu mp lose suction due to roll or pitch of the vessel. The oil shall be properly strained and cooled before being delivered to the parts to be lubricated, Strainer (if provided) shall be fitted on the discharge side of the lubricating oil pump. Satisfactory means of cooling the oil shall be provided. Oil coolers, if provided, shall conform to type A of Specif ication MIL-C-15730. Coolers shall be class 2 unless otherwise specified (see 6. 1). The coolant side shall be given a hydrostatic test of 225 p. s. i. g. unless otherwi se specified (see 6. 1). Suitable means shall be provided for regulating the pressure of the lubricating oil. The unit shall be designed to operate in accordance with Specification MIL- L-17331.

3.3.12 Bedplate. - The bedplate shall be continuous under the compressor and its driving unit. It shall be machined to receive the Intermediate frames or housing, or may be integral therwith. All top and bottom bearing surfaces shall be machine finished. Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship, without disturbing the alignment of the assembled units, and such that the normal distortion, weaving or vibration of the supporting structures onboard ships cannot cause misalignment of components

3.3.12.1 The basis of size 10 compressors shall have six 3/4 inch holes for 1 /8-inch foundation: bolts, arranged as shown on figure 1.

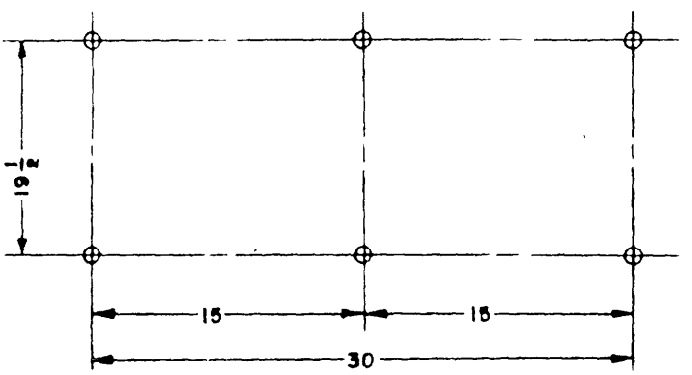


Figure 1.- Size 10 compressors

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3.3.13 Housing.- The housing shall consist of cast frames, cast frames and forged tie rods, or structural welded assembly which shall support the cylinders. The crankpits and the housing shall be wholly enclosed to prevent the splashing of oil. Access openings with covers shall be provided for observation and adjustment of the bearings, and where necessary breather openings fitted with screens shall be provided.

3.3.14 Air valves.- The air valves shall be of the thin plate type of an efficient and durable design. Air valves shall be readily accessible for inspection and overhaul.

3.3.15 Cooling system.-

3.3.15.1 The compressor shall be provided with an efficient water cooling system, including water jacketing for cylinders and heads, intercoolers and aftercooler. The temperature of air delivered from the aftercooler shall not exceed the inlet temperature of the circulating water by more than 15° F. The maximum temperature of air discharged from any stage of compression to the cooler following shall not exceed 400° F. under any normal operating condition.

3.3.15.1.1 Consideration will be given to air cooling, in lieu of water cooling, of cylinders providing all other specified indications are met, providing the additional heat given off to the air does not exceed 32 b. t. u. per minute while the machine is running, and providing it is shown that there will be a saving in weight and cast thereby.

3.315.2 All parts of the cooling system shall be of corrosion-resisting material and shall be designed for the use of salt water supplied from the ship's flushing or fire mains, or, when specified, from an attached circulating water pump. The cooling system shall be designed for a working pressure of 50 p.s. i. at the cooling water relet and shall be capable of withstanding a maximum pressure of 150 p.s.i. Provision shall be made for completely draining all parts of the circulating water system.

3.3.15.3 The cooling water system shall be so arranged that the quantity of cooling water supplied to the oil cooler (if provided) may be regulated without affecting the quantity of cooling water supplied to the remainder of the system.

3.3. 15.4 Provision shall be made for positive and automatic control of the cooling water so that the water is turned on when the compressor is running and turned off when the compressor is not running (see 3. 5.3. 1).

3.3. 15. 5 Coolers inter and after.- Inter and after coolers shall be of the straight-tube type, Consideration will be given to the use of coil-type coolers. Above pressures of 250 p.s.i.g. , the air shall flow through the cooler tubes, with water flowing over and around the tubes. Cooler tubes shall be rolled or welded into tube sheets or headers at both ends. Provision shall be made for the expansion of tube nests. Stay rods, if and as necessary shall be provided for support of tube sheets and air heads. Cooling coils, if used, shall be rigidly tied together by strips brazed to each coil turn. The complete coil assembly shall be supported in the cooler in a manner that will prevent vibration of the coil. Each cooler tube nest shall be a self-contained unit so that it may be easily removed from the compressor as a unit for inspection and cleaning. Suitable water and air relief valves shall be fitted on each cooler, except that a water relief valve on the first intercooler will not be requires. Unless otherwise specified in the contract or order water relief valves shall be set at 155 p. s. i.

3.3.15.6 Moisture separators.- Efficient separators shall be provided for removing free moisture and oil from the air upon delivery from each intercooler and aftercooler, All separators shall be fitted for blowing out accumulations of water and oil. Drain provisions shall be accessible for operation from the operating or- gage board side of the unit, insofar as practicable.

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3.3.15.7 Zincs.- The upstream side of the inter and after coolers shall be fitted with zincs of a design similar to any one of those shown on Drawing 5000-S4823-841055. The zincs shall have as large a surface area as is practicable for maximum electrolytic protection without causing excessive turbulence. The zincs shall be easily removable for inspection and replacement. Zincs shall De special high grade material as specified in Specification MIL-A-18001.

3.3.16 Unloading. - All compressors shall be fitted with automatic unloading valves so that the compressor will always be started in the completely unloaded condition, that is, no load except inertia and friction load of the machine (see 3.5.3. 3),

3.3.17 Fittings.-

3.3.17. 1 Gages and gage board.- All gages shall be of the 3-1 /2-inch size and as shown in Specification 45G1. All gages, instruments, and similar items, shall be mounted. on a common gage board. The gage board shall be provided with brackets for m ounting on the compressor, Mounting brackets shall be of a type which can be removed from the compressor when the gage board is bulkhead mounted. Pressure gages shall be provided to indicate the air-delivery pressure from each stage of compress ion, the pressure in the circulation water system and the lubricating oil pressure.. All gages shall be suitably identified. The air gages for each stage shall be marked to show the normal designed working pressure, and such working pressures shall be indicated on drawings technical manuals.

3.3.17.2 Thermometers.- Plugs shall be provided in the intake and discharge air channels of each stage, and in the delivery line beyond the aftercooler, for the insertion of thermometers. Similar plugs shall also be fitted at the inlet and discharge of the circulating water system and in the oil reservoir. A complete set of thermometers and thermometer sockets shall be furnished with each machine.

3.3.17.3 Final check valve. - The final air discharge shall be provided with a check valve and a shut-off valve.

3.3.17.4 Air filters. - Compressor intakes shall be provided with air filters of a type that will prevent the entrance of injurious substances and that will reduce the noise resulting from suction pulsations.

3.3.18 Piping and valves. - All water, air and oil connections to the compressor shall be readily accessible. Pockets where oil or water may collect shall be avoided, or if unavoidable, shall be arranged for convenient drainage. All compressor water connections to external shipboard piping systems shall be flanged. Flange and bolting standards shall be in accordance with Drawing B-814. All root connections shall be in accordance with Drawing 5000-S4823-F-841337. All air, water and oil piping on the unit shall be made up with connections so as to be readily removable for overhaul of the unit. Threaded pipe connections shall be reduced to an absolute minimum on internal piping. Compression type silver-brazing or flanged fittings shall be used wherever possible. Piping shall be so fitted as to avoid necessity of removing any part of same for access to and removal of crankcase and valve covers. All relief valves shall have the discharge directed in such manner as not to discharge in the face of operating personnel.

3.3.19 Threaded parts. - All threaded parts shall be in accordance with Handbook H28. Use of studs shall be avoided.

3.3. 20 Jacking device. - Suitable provision shall be made for jacking the compressor over, if it is impractical to turn the machine by hand.

3.3.21 Welding and allied processes. - Welding and allied processes shall be in accordance with Section S1-4.

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3.3.22 Painting.- All surfs-es to be painted shall be thoroughly cleaned and free from oil, grease, mill scale, rust an loose welding beads, All external nonworking ferrous and aluminum surfaces shall be painted with one coat of zinc chromate primer, in accordance with Specification JAN- P-735, and with two coats of haze gray paint in accordance with Specification MIL-P-15130.

3.3.22 Identification plates.-

3.3.22.1 Identification plates shall be furnished. on each compressor and shall be type A or B in accordance with Specification MIL-I- 15024, except that identification plates of plastic or aluminum are not acceptable.

3.3. 23. 2 Identification plates shall be secured to equipment with corrosion resistant metallic screws.

3.3.23.3 Identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Service application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
  (1) Capacity.
  (2) Pressure in pounds per square inch.
  (3) Speed Of shaft in revolutions per minute.
  (4) Brake horsepower.
  (5) Working pressure at each stage.
  (6) Lubricating oil pressure.
  (7) Cooling water pressure required.
  (8) Set pressures at each relief valve.
  (9) Any special data vital to the unit,
(f) Contract number (and item number for multiple unit orders).
(9) Bureau of Ships stock number.
(h) Section for inspector's stamp,

3.3. 23.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.4 Materials.-

2. 4.1 All material used in the construction of the compressors and associated equipment shall be *of* the quality best suited for the purpose intended, and shall be strictly in accordance with the material specifications as listed in table I.

3.4.2 The use of materials differing in properties from those specified for a given application will be considered when the contractor shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to the bureau or agency concerned.

3.4.3 Where no specifications are specified herein for the material for certain parts, the material used shall be of a type and quality consistent with that specified for associated parts in the compressor. The use or substitution of manufacturer's material shall be subject to approval by the bureau or agency concerned and shall be clearly indicated in a list submitted with the master drawings.

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3.4.4 When the contract or order specifies that the compressor units shall **be** nonmagnetic the materials of construction shall be selected such that the magnetic permeability of the compressor component unit (less motor and controller) after fabrication will be 2.0 or less measured by a permeability indicator conforming to Specification MIL-I-17214.

3.4.4.1 In order to achieve the desired low permea bility after fabrication it is advisable to choose materials which have a permeability prior to fabrication of 1. 2 or less when materials whose permeability} is affected by fabrication (CRES) are used.

Table I - Material.

Application	Material	Specification
Bedplates	Cast steel Structural steel, welded Corrosion resisting steel Aluminum bronze	MIL-S-15083 or MIL-S-17509 MIL-S-20166; MIL-S-16113 QQ-S-763; QQ-S-766 or MIL-S-17996 MIL-B-15939
Frames and housings	Cast steel Structural steel, welded Nodular graphitic iron Gun metal Valve bronze	MIL-S-15083 or MIL-S-17509 MIL-S-20166; MIL-S-16113 MIL-I-17166 MIL-M-16576 MIL-B-16541
Cylinders, air; Cylinder heads	Gun metal Valve bronze see note 2	MIL-M-16576 MIL-B-16541
Cylinder liners	Hard close grained cast iron	MIL-G-858, QQ-I-652
Crankshaft; Connecting rods, caps, and bolts	Forged steel, class As or Bs Corrosion testing steel Aluminum bronze	MIL-S-890 QQ-S-763 or MIL-S-17996 MIL-B-15933
Wrist pins	Forged steel, class Ac or An, hardened and ground Corrosion resisting steel	MIL-S-890 QQ-S-763 or MIL-S-17996
Wrist pin bushings	Bearing bronze, grade I or II	MIL-B-16261
Pistons	High test gray iron Cast iron Aluminum	QQ-I-652 MIL-A-958

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Table I- Materlal (cont'd.),

Application	Material	Specification
Piston rings	Hard close grained cast iron centrifugally cast	QQ-I-652
Air valves	Corrosion-resisting steel, class FS-304 or FS-410	QQ-S-763 or MIL-S-17996
Air valve seats	Corrosion-resisting steel, class FS-304 or FS-302 Corrosion-resisting steel, class 1 or 2, respectively Bronze Bronze	QQ-S-763 or MIL-S-17996 MIL-S-867 or MIL-S-17509 MIL-B-16540 MIL-B-892
Air valve springs	Spring steel, cadmium plated Corrosion-resisting steel	JAN-S-1160 QQ-S-763 or MIL-S-17996
Intercooler Tubes, seamless	See note 1 Copper nickel alloy Nickel copper alloy	 MIL-T-15005 MIL-T-1368
Intercooler tube sheets or headers	Naval brass Nickel copper alloy	MIL-B-994 MIL-N-894
Intercooler Air heads and moisture separators	Aluminum bronze, class 1 Gun metal Valve bronze	MIL-B-16033 MIL-M-16576 MIL-B-16541
Intercooler Waterheads	Gun metal Valve bronze	MIL-M-16576 MIL-B-16541
Intercooler sheels	Seamless drawn tubing same as tubes, or Gun metal Valve bronze	MIL-T-16420 MIL-M-16576 MIL-B-16541
Intercooler shell bottles	Naval brass, or same material as shell CU-NU	MIL-B-994 MIL-C-15726

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Table I - Material (cont'd.).

Application	Material	Specification
Intercooler	Naval brass	MIL-B-894
Stay rods	Nickel copper alloy	MIL-N-894
Intercooler gaskets (air side)	Copper, soft iron, or dead soft aluminum sheet	QQ-C-576 QQ-A-561
Sheaves	Cast steel	MIL-S-15083 or MIL-S-17105
	Cast iron	MIL-G-858
	Valve bronze	MIL-B-16541
	Nodular graphitic iron	MIL-I-17166

Notes. -
1. Seamless tubes for intercoolers shall be 90-10 copper-nickel. Pressure parts subject to pressures in excess of 500 p. s. i. g. , shall be of 70/30 copper-nickel or nickel-copper alloy.
2. Air cooled cylinders may be of modular graphitic iron in accordance with Specification MIL-I-17166.

3.5 Electrical equipment.-

3.5.1 General.- All electrical equipment, for w hich specifications are not specifically set forth herein, shall conform to Specification MI L- E- 917. The degree of enclosure for electrical equipment shall be watertight unless other enclosures are specifically specified herein.

3.5.2 Motors. - All motors shall be in accordance with Specification MIL-M-17413 for direct-current (d. c.) motors, or Specification MIL-M-17060 for alternating current (a. c.) motors, and the following:

(a) Service	A
(b) Voltage and current characteristics	As specified (see 6. 1)
(c) Enclosure	Dripproof protected
(d) Mounting	Horizontal
(e) Ambient temperature	50° c.
(f) Duty	Intermittent - Rum 2 minutes out of every 3 minutes for 1 hour
(g) Speed	Constant
(h) Winding	D. c. - stabilized shunt A. c, - squirrel cage induction
(i) Insulation	class A, unless otherwise specified in the contract or order
(j) Bearings	Ball

3.5.2.1 The horsepower rating of each motor shall be in accordance with Navy standard motor ratings, but not less than 1.05 times the brake horsepower of the driven compressor under rated conditions of capacity and pressure.

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3.5.3 Controllers.- Controllers shall be in accordance with Specification MIL-C-2212, for a. c. controllers or Specification MIL-C-2174, for d. c. controllers and the following:

Duty	Intermittent- Run 2 minutes out of every 3 minutes for 1 hour
Enclosure	Dripproof
Operation	Magnetic
Ambient temperature	50° c.
Control function	Motor starting
Type	A. c. - Across the line D. C. - Resist.er
Protection	Thermal overload, low voltage protection
Insulation	Class A (coils only) and B
Performance	Automatic (see 3. 5. 3.1 and 3.5.3,2)
Master switches	Remote pressure switch; and local pushbuttons (see 3,3.15.4)

3.5.3.1 Unless otherwise specified in the contract or order the performance shall be automatic. It shall be necessary to push the "start" button to put the motor under automatic control of the pressure switch; and the motors shall then remain under automatic control until the. "stop" button is pushed. If the motor should stop due to overload or low voltage, it shall not restart until the "start" button is again pushed.

3.5.3.2 When specifically required by the contract or order controllers shall provide both semi-automatic and automatic performance. In this case a 2-position switch shall be provided marked ' 'constant speed" and "start-stop".

3.5.3.2.1 Under the "constant speed" control the performance shall be semi-automatic under control of a two element pushbutton station marked "On", and "Off". No "emergency run" pushbutton shall be provided.

3.5.3.2.2 Under the "start-stop" control the performance shall be automatic under control of the pressure switch.

3.5.3.3 The pressure switch shall be in accordance with Specification MIL-C-2174 or MIL-C-2212, as applicable and shall be watertight, The pressure switch shall be of an adjustable type, suitable for at least 300 to 600 p. s. 1. adjustment and shall be set to start and stop the compressor between the lim its of 550 and 600 p. s.i. g.

3.5.3.4 Cooling system valves (see 3.3.15.4), if electrically operated, and unloading valves (see 3.3.16), if electrically operated, shall conform to all the applicable requirements for controllers as specified in 3. 5.3, except that enclosures shall be watertight (or better). Valves shall be tight-seated and all internal parts in contact with water shall be of nonferrous material except where ferrous parts are necessary for the magnetic circuit, and such parts shall be suitably nickel plated for protection against corrosion. Cooling system valves shall be provided with a means of manually opening the valves in case of coil failures (see 3.3. 18).

3.6 Repair parts and tools. -

3.6, 1 Onboard.- Repair parts shall be furnished and processed in accordance with Specification MIL-R-15137.

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3.6.2 Each set of repair parts and tools required by the contract or order shall contain the parts listed in table H, plus any additional parts or tools considered by the bureau or agency concerned or by the manufacturer to be essential in successful servicing of the compressors. Unless otherwise specified in the contract or order, the quantities of repair parts per set shall be based on one compressor per ship (that is, one compressor per set of repair parts).

3.6.3 Repair parts for lubricating oil coolers, motors, controllers, pressure Switches and other electrical devices shall be furnished as required by the applicable specifications listed in 2.1,

3.6.4 Tools.-

3.5.4.1 Special tools and wrenches.- A complete set of special tools and wrenches such as inter-cooler test rings, piston ring compressors, wrenching plates, and other tools and wrenches necessary for the proper repair and maintenance of the compressors shall be furnished for each compressor. The tools and wrenches snail be marked for the purpose intended. The tools shall be stored in separate steel boxes.

Table II - Onboard repair parts.

A - 100 percent complete replacement for all compressors installed per ship:	
Bearings, ball	Springs, valve, inlet and discharge
Gaskets, complete sets	Swabs, lubricating, ring type, for piston rod
Rings, piston, for all stages	Valves, discharge, for all stages
Rods, connecting, if not bushed	Valves, inlet, for all stages
B - 200 percent complete replacement for all compressors installed per ship:	
Belts, "V"-type, complete	
C - 100 percent for one compressor:	
Bearings, crank pin	Pins, wrist or crosshead
Bearings, main	Piston, air, rod and nut
Bearings, wrist pin or crosshead pin	Pumps, oil, complete
Bolts, nuts, washers, and similar items, connecting rod (crank end)	Springs, automatic control device
	Springs, relief valves
	Strainers, oil
	Thermometers
Bolts, nuts, washers, and similar items, connecting rod (crosshead)	Tubes, inter- and after-coolers
	Valves, discharge for all stages, complete with seats, guards, springs, and similar items
Bushings, including thread bushings	
Coils, electrical, one of each size and type	Valves, inlet, for all stages, complete with seats, guards, springs, and similar items
Couplings, flexible	Working parts for automatic control devices
Gages, oil level	
Gages, pressure	
D - 200 percent for one compressor, as listed under C above, shall be furnished if three or more compressors are furnished per ship.	

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3.6.4.2 When the contract or order specifies non-magnetic equipment, all special tools shall be of non-magnetic material (see 3.4. 4).

3.6.4.3 Tools which are available in the Catalog of Naval Material, General Stores Section, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools (special tools are defined as those tools not listed in the Catalog of Naval Material, General Stores Section (copies of this catalog may be consulted in the office of the Government inspector)) shall be detailed, and included in the list of material.

3.6.4.4 Each box containing onboard repair parts shall contain a list entitled “List of onboard repair parts anti tools”, The list shall be in format suitable for use on outline drawings and in the instruction books. It shall not be less than nominal 8-1/2 by 11 inches in size.

3.6.4.4.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacturer _____
Quantities are for _____ units per ship
The columns shall include:
Piece number _____
Name of part _____
Quantity _____
Drawing number _____
Standard Navy stock number _____
Additional columns may be used as applicable

3.6.4.4.2 The list shall be so treated as to be resistant to oil, water and fading.

3.6.5 Stock.- The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with Specification MIL-R-15137.

3.6, 5.1 The cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania, unless otherwise specified in the contract or order.

3.7 Drawings.-

3.7.1 Drawings shall be in accordance with type I of Specification MIL-D-963.

3.7.2 Outline drawings shall be furnished covering complete units including compressor, driver, bedplate and attached accessories. The center of gravity of complete units shall be indicated on each outline drawing. Separate outline drawings shall be furnished for each contract or order and for each different unit on a contract or order.

3. 7.3 Compressor master drawings shall include sectional assembly, complete list of material and details of all parts. The sectional assembly and list of material with references to detail drawings should be on a single sheet; and details shall be on one or more additional sheets. Several details should be shown on each master drawing, Details of single parts on size A sheets, resulting in a multiplicity o drawing sheets, should be avoided. Size F drawings (28 by 40 inches) are preferred.

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3.7.4 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing.

3.7.5 Drawings for drivers and associated equipment shall be in accordance with the specifications for those equipments.

3.7.6 The sectional assembly drawings shall show all assembl y clearances, and snail reference the qualification test, high-impact shock test, and inclination test reports, and the bureau or agency letters approving those reports.

3.1. 7 Drawings submitted with bids are not required to be in master drawing format.

3.8 Technical manuals.- Technical manuals shall be in accordance with type C of Specification MIL-T- 15071, unless otherwise specified in the contract or order, A separate technical manual shall be furnished for each different compressor unit except in special cases when books covering more than one compressor may be approved. Preliminary technical manuals submitted for approval snail include all proposed sections complete.

3.8.1 Illustrations.- Two photographs, taken 180 degrees apart on a horizontal plane shall be included. Reduced size reproductions of approved outline, assembly and sectional plans of the complete unit, and reduced size reproductions of approved electrical plans and certification data shall also be included. Approved plan excerpts and exploded views of parts or assemblies requiring detailed instruc - tion, such as valve assemblies, unloading mechanisms, snail be included.

3.8.2 Distribution.- Two copies of technical manuals shall be packed with each compressor unit. Additional copies of technical manuals shall be furnished as specified (see 6. 1). If final technical manuals are not available at the time compressors are shipped then two copies of preliminary technical manuals shall be packed with each compressor unit; and the final manuals shall be delivered at a later time as required b: Specification MIL-T-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests at a Government la boratory. - Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4. 4.

4. 1.1 Consideration will be given to granting qualification as a result of service tests. To be granted qualifications as the result of service tests the following requirements shall be met:

(a) Two or more compressors of the general design and general operating characteristics shall have given satisfactory service onboard a Naval vessel for a period of at least two years and shall still be in service on such vessel at the time qualification is requested.

(b) When the design of a compressor in service for which a manufacturer requests qualification does not meet current Naval requirements as regards pressure and capacity characteristics, the manufacturer shall be required to furnish satisfactory evidence that he has produced compressors of design characteristics equal to those currently required for Naval service. The Bureau of Ships reserves the right to require additional qualification tests of any design of compressor, which, in its opinion, has been changed sufficiently to require further demonstration of Its suitability.

(c) Any manufacturer whose product has been eligible to be placed on the qualified products list as a result of service test shall be required on his first contract or order after being granted qualification to submit a complete unit to the shock test. The awarding of qualification shall be contingent upon the unit passing the shock test.

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4.2 Sampling.- Completely assembled compressors, including driving motors, controllers, and. a ll equipment or fittings shall be selected by the Government inspector in accordance with table III,

Table III - Sampling for tests.

Number of compressors on contract or order	Number of compressors to be tested
1 to 5	1
6 to 8	2
9 to 15	3
16 to 25	5
26 to 65	7
66 to 110	10
111 to 190	12

4.3 Tests at the place of manufacture. -

4.3.1 The various parts of each compressor shall be hydrostatically tested at the place of manufacture in accordance with the following:

(a) Cylinders, 1-1/2 times the maximum designed working pressure.
(b) Water jackets, 225 p.s. i. g.
(c) Intercoolers, and aftercoolers, 225 p. s, i. g. on the water side and 1-1/2 times the maximum design working pressure on the air side,
(d) Piping and other parts not mentioned above and subjected to pressure shall be tested to 1 -1/2 times the maximum designed working pressure.

4.3.2 Each unit selected in accordance with 4.2 shall be subjected to the tests specified in 4

4.2.3 Each compressor not selected in accordance with 4.2 shall be subjected to the tests specified in 4.4.3

4. 3.4 Test method.- Capacity tests of compressors shall be in accordance with the test procedures outlined in the second edition of the Compressed Air Handbook published by the Compressed Air and Gas Institute.

4.4 Test procedures.-

4.4.1 Electrical.- Motors and controllers shall be tested as required by the applicable specifications listed in section 2.

4.4.2 Performance tests. - Compressor units selected in accordance with 4.2 shall be submitted to a continuous running test of 8 hours' duration at normal capacity and pressure. Test shall run under normal electrical conditions specified in the contract or order, Power consumption tests shall be conducted as a part of the tests. On these tests the following data shall be recorded at the start and finish of the run and at regular intervals, not exceeding 1/2 hour, throughout the run:

4.4.2.1 Recorded data. -

MECHANICAL

(a) Dry bulb ambient temperature.
(b) Wet bulb ambient temperature.
(c) Intake temperature (in air stream close to intake).
(d) Barometer readings.

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(e) Air discharge pressure from each stage of compression,
(f) Final air pressure at discharge from after-cooler.
(g) Air temperatures before and after each stage of compression.
(h) Circulating water inlet and outlet temperatures.
(i) Quantity of cooling water in gallons per minute.
(j) Revolutions per minute of driving unit and of compressor.

ELECTRICAL

(a) Voltage, amperage, and frequency.
(b) Power factor.
(c) Power input to motor.
(d) Complete data as to performance of motor required to determine suitability of the electrical equipment for the application. This data shall be recorded primarily to demonstrate that the motor is of adequate power for the service intended.

OPERATIONAL

(a) Operation of the automatic temperature control.
(b) Operation of the automatic unloading devices.
(c) Operation of the automatic stop-start control. The ability of the automatic regulator to stop and start the compressor in accordance with the requirements of this specification shall be demonstrated for each unit. On these tests the automatic regulator shall be adjusted to function at the maximum, minimum, and mean pressures included in the range of adjustment specified.
(d) Operation of moisture separators. (Include record of quantity of oil and water drained from each separator.)

4.4.2.1.1 Computed and recorded data. - The following data shall be computed from the recorded data and included on each test data record:

(a) Capacity in cubic feet of free air per minute actually delivered.
(b) Brake horsepower at the compressor shaft (computed from previous test of motor),
(c) Volumetric efficiency.
(d) Overall efficiency of compressor (that is, compression efficiency X mechanical efficiency). Theoretical air horsepower upon which efficiency is based may be calculated from either the adiabatic or isothermal base, but test data should indicate which is used.

4. 4.2.2 During each test the functioning of the compressor units shall be smooth and free from excessive heating, noise, and vibration, and the performance in all respects shall meet the requirements of this specification..

4.4.2.3 Upon completion of the performance tests, each compressor shall be disassembled sufficiently to permit inspection of all parts subject to wear or damage by running, these parts shall include bearings, wrist pins, pistons and piston rings, cylinders, and air valves. To be acceptable no parts of the compressors shall show any signs of failure or excessive wear. The condition of the compressors at the conclusion of all tests shall be noted on the official test sheets.

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4.4.3 Each compressor unit not selected in accordance with 4.2 shall be subjected to tests exactly as required by 4.4.2, except:

(a) The duration of the test need not exceed 4 hours, and data need be taken only at the start and finish of the run and arc in between,
(b) The units need not be disassembled upon completion of the -performance tests, unless excessive nesting, noise, vibration, or erratic performance is noted during the tests.

4.4.3.1 If excessive heating, noise, vibration, or erratic performance is noted, and is corrected the 4-hour test shall be repeated. If the first correction does not completely eliminate the trouble, then after further correction the compressor snail be subjected to a complete 8-hour test before acceptance.

4.4.3.3 All record d data and co ip nations required ior the 8-hour test are also required for the 4-hour test.

4. 4.4 Inclined operation. - One compressor unit, including the driving motor, shall be selected by the Government inspector for tests to determine satisfactory compliance with 3.3.1.

4.4.4.1 Only one compressor of each design shall be subjected to the inclination tests, and the tests need not be repeated on later contracts or orders unless the Government inspector or bureau or agency concerned considers that sufficient changes have been made from the tested design to warrant retesting. Neither changes in materials only of compressor construction, nor changes in the driving motor shall be considered as reasons for retesting.

4.4.4.2 The -compressor unit shall be operated at rated load for not less than 30 minutes with the base tilted 15 degrees in each of the four cardinal positions, right, left, f rent, and rear. During the progress of these tests there shall be no excessive heating of any parts, and noise and vibration shall not exceed that of normal horizontal operation.

4.4.4.2 The suitability of the compressor unit for operation when a ship is rolling or pitching shall be determined by inspection of the design.

4.4.5 High-impact shock test. - One compressor unit including the driving motor, shall be selected by the Government inspector for type A shock tests in accordance with Specification MIL-S-901.

4.4.5.1 Only one compressor of each design shall be subjected to the shock tests, and the tests need not be repeated on later contracts or orders unless the Government inspector or bureau or agency concerned considers that sufficient changes have been made from the tested design to warrant retesting. Changes in materials of construction of the compressor will require retesting. Changes in the driving motor shall not be considered a reason for retesting of the compressor.

4. 4.5.2 The compressor shall be idle and running on alternate blows of the shock test, but the compressor shall not be compressing air while runnihg. .

4. 4. 5.3 Inspection after shock test.-

4. 4.5.3.1 Motor. - Inspection of the motor shall be in accordance with Specification MIL-M-17060 or MIL-M-17413.

4.4.5.3.2 Compressor. - Inspection of the compressor shall be conducted as follows:

(a) Examine the unit carefully and note the extent of any external damage,
(b) Jack the compressor over several times by hand to insure that all working parts are free and ready for starting,
(c) Run the compressor unloaded for a short time and note any mechanical peculiarities,
(d) Gradually build up the had and carefully check any undue noises, excessive vibration.
(e) Secure the compressor and disassemble for complete inspection. Observe and record all effects of the shock test on the various compressor components.

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4.4.5.3.3 The motor shall be considered to have failed the shock test unless it conforms to Specification MIL-M-17060 or MIL-M-17413.

4.4.5.3.4 The compressor shall be considered to have failed the shock test in the event of the following:

(a) Brea kage of all parts, including mounting, bolts, and similar items.
(b) Detachment of any part or object from the unit.
(c) Compressor' requires attention or adjustment or replace ment of parts to enable it to perform its principal function during and after each, blow of the test.
(d) Unit falls to function satisfactorily as specifically defined in this specification

4.4.5.4 Disposition of shock-tested equipment. - Equipment which has successfully pushed the shock tests will be considered acceptable for such service as designated by the bureau or agency con- - cerned. The contractor shall thoroughly recondition such units, correcting any min or imperfections or deformations, and shall place the equipment in satisfactory working order. The motors shall be recon - ditioned and tested in accordance with the requirements of the motor specifications. After recondition- ing, the complete equipment shall be tested as required by 4.4.3.

4.4.5.4.1 Equipment which has been subjected to the high-impact shock test and has failed will not be accepted, either in whole or in part. However, failure of the compressor only shall not cause rejection of an otherwise acceptable motor, nor shall failure of a motor only cause rejection of an other- wise acceptable compressor,

4.4.5.5 If a manufacturer is unable to conduct the shock tests at his own plant, he may arrange to have them conducted at a commercial or Government laboratory suitably equipped to conduct the tests. When the shock tests are to be conducted at a Government laboratory a complete set of master and outline drawings shall accompany the equipment.

4.4.6 Test reports. -

4.4.6.1 Test reports shall be prepared for all tests conducted by the contractor, including the shock and inclination tests. Recorded data of the compressor performance tests may be logged on contractor's forms, The following identification data for each compressor unit shall be included in the test report:

(a) Serial numbers of the compressor, driving unit, and control.
(b) Ship vessel for which intended,
(c) Contract or order number,
(d) Actual net weight of the unit in the "dry" and “wet” condition.

4.4.6.2 Distribution of compressor performance test reports shall be as follows:

(a) Two to the bureau or agent] concerned.
(b) Two to each Supervisor of Shipbuilding for each ship's worth of compressors furnished (one for supervisor's files; balance for shipbuilder) for privately built Ships.
(c) Two to each Naval shipyard for each ship's-worth of compressors for Navy built ships.
(d) One for primary inspection officer.
(e) One for branceh inspection office (if any).

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4.4.6.3 Distribution of shock test reports and inclination test reports shall be as follows:

(a) Six to the bureau or agency concerned, for approval.
(b) One for the primary inspection office.
(c) One for branch inspection office (if any).

4.3 Inspection procedures, - For Naval purchases, the general Inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5. 1 Equipment and repair parts shall be prepared for delivery in accordance with the following requirements as specified (see 6. 1):

(a) For domestic shipment - immediate use.
(b) For domestic shipment and storage.
(c) For overseas shipment.

5.2 Preservation am! packaging. -

5.2.1 For domestic shipment - immediate use. - The compressor shall be cleaned, preserved and packaged in accordance with commercial practice,

5.2.2 For domestic shipment and storage or overseas Shipment, - Cleaning, preservation and packaging methods and materials shall be in accordance with Specif ication MIL- P- 116.

5. 2.2. 1 Compressor. - Internal surfaces of the compressor shall be flushed with preservative type P-10. Exterior, unpainted surfaces subject to corrosion shall be coated with preservative type P-2 and wrapped or covered with greaseproof barrier material conforming to grade A of Specifica- tier, JAN-B-121 secured in place with tape as necessary, Openings to the interior of the compressor shall be sealed with tape conforming to grade B of Specification JAN-P-127 or barrier material conforming to Specification JAN-B-121 secured in place with tape.

5.2.2.2 Inter- and after-coolers. - Intercoolers and aftercoolers shall be dried by passing dry air through the system. All openings in the coolers shall be sealed.

5.2.2.3 Motor. - The motor shall be preserved and packaged in accordance with method II of Specification MIL-P-116. If the mounting does not permit obtaining a water-vaporproof seal as required in method II, the alternate method of sealing the motor as specified in Specification MIL-P-1629e may be used.

5.2.2.4 Reduction gears. - Reduction gears shall be flush with preservative type P-2.

5.2.2.5 Instruments panel board. - Instruments not removed from the basic unit shall have all openings sealed with pressure sensitive tape conforming to type I, grade B of Specification JAN-P-127. Unless otherwise protected against breakage, dial glasses shall be masked with masking tape, covered with barrier material, in accordance with type I, grade A, class 1 of Specification JAN-B-121, then protected with pieces of plywood of fiberboard secured in place with pressure sensitive tape.

5.2.2.6 Controllers. - Detached controllers and other detached electrical components shall be preserved and packaged in accordance with Specifications MIL-P-16018 and MIL-P-17652, as applicable.

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5. 2. 3 Repair parts and tools .-

5.2.3.1 Electrical repair parts and took shall be preserved and unit packaged in accordance with Specification MI L- P-17652 and MIL-P-16298, and in accordance with Specification MI L- P-116 for items not specifically covered therein

5. 2. 3. 2 Mechanical repair parts shall be preserved and unit packaged in accordance with Publication JPI-6 and in accordance with Specification MI L-P-116 for items not specifically covered therein.

5. 2. 3. 3 When specified in the contract or order, onboard repair parts shall be packaged as a set in onboard repair parts stowage boxes conforming to type M or W of Specification MIL-B-233 as specified. The hardware of type W shall be of non-magnetic materials exclusively. The gross weight of the boxes shall not exceed 200 pounds.

5.3 Packing. -

5. 3. 1 Equipment - for domestic shipment - immediate use. -

5. 3. 1.1 The equipment and related accessories (except repair parts and tools) packaged as specified in 5. 2 . 1 shall be packed in a manner which will insure acceptance and safe delivery at destination. The containers shall comply with the Consolidated Freight Classification Rules or Motor Freight Class ification Rules, whichever may be applicable.

5.3.2 Equipment - for domestic shipment and storage or overseas shipment. -

5.3.2.1 Components removed from the basic unit to facilitate shipment, which require segregation or protection from mechanical damage, shall be packaged in nailed wood boxes conforming to Specification NN-B-621 or in nailed wood cleated boxes conforming to Specification PPP-B-601.

5. 3. 2. 2 The complete unit, including detached components, shall be anchored to the base of a nailed wood sheathed crate conforming to Specification MIL-C-3769 or JAN-P-104, as applicable. Blocking, bracing and cushioning shall be in accordance with Specification JAN-P- 100, The crates shall be ventilated in the manner specified in Specification JAN-P-104.

5. 3. 2. 3 Repair parts and tools. - Repair parts and **tools** shall be packed for domestic shipment and storage or overseas shipment as specified (see 6. 1) in accordance with Specification MIL-P-17652.

5. 4 Publications. - Technical manuals shall be packaged and packed in accordance with Specification MIL-T-15071.

5. 5 Drawings. - Drawings shall be prepared for shipment in accordance with Specifi - cation MIL-D-963.

5. 6. Marking. - In addition to any special marking required in the contract or order, marking of the packages and shipping conta ners shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1. 2).
(c) Characteristics of electrical supply.
(d) Operating conditions of the compressor (see 3.3. 2).

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(e) Class of cooler requi red and when test pressure if other than 225 p.s. i.g. (see 3.3. 11).
(f) Electrical character sties (see 3. 5).
(g) If other than automatic operation is required (see 3.5,3. 2).
(h) When additional technical manuals should be furnished (see 3.8. 2).
(i) W hether the corr.presser unit shall be prepared for domestic shipment - immediate use; domestic shipment and storage; or overseas shipment (see 5.2 and 5. 3).
(j) Whether onboard repair parts boxes are required; if so, specify type M or (see 5,2.3).
(k) Whether the repair parts shall be prepared for domestic shipment and storage or overseas shipment (see 5. 3. 2. 3).

6.2 Qualification. - In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Qualified Products List. The attention of suppliers is called to this requirement, and ma nufacturers are urged to arrange to have the products that they propose *to* offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.3 Lubricating oil coolers for submarine applications should be class 1.

6.4 Superseding data. - This specification supersedes that part of Section S49-2 of the General Specifications for Machinery covering class T compressors.

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise **as** in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto,

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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